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Eco-Epidemiology and Genotype Susceptibility of Banana Scarring Beetle (*Basilepta subcostatum*) in Saharsa, Bihar

Anshu Prabha & Arun Kumar*
University Department of Zoology, B. N.M. University, Madhepura, Bihar, India

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Abstract- The banana leaf and fruit scarring beetle (*Basilepta subcostatum* Jacoby) has emerged as a significant pest threat in the banana-growing regions of North Bihar, particularly in Saharsa. This study investigates the seasonal population dynamics, host plant susceptibility, and environmental correlates of *B. subcostatum* over a one-year period (April 2022-March 2023) under natural field conditions. Two genotypes were evaluated: the local cultivar *Musa paradisiaca* cv. 'Malbhog' and the wild species *Musa balbisiana*. Results revealed a marked post-monsoon peak in beetle population, with 'Malbhog' showing significantly higher infestation levels and damage compared to the more tolerant *M. balbisiana*. Adult beetles caused severe scarring on fruits and leaves of 'Malbhog', reducing commercial value, while the wild genotype sustained minimal injury. Statistical analysis indicated a strong positive correlation between beetle abundance and relative humidity (r = 0.72), and a moderate correlation with rainfall (r = 0.58), while high temperatures (>35°C) negatively influenced beetle activity (r = -0.45). These findings emphasize the importance of integrating resistant genotypes, seasonal monitoring, and climate-informed interventions to develop sustainable pest management strategies. The study also highlights *M. balbisiana* as a valuable genetic resource for breeding beetle-resistant banana varieties.

Key words: ***Basilepta subcostatum, Musa balbisiana,*** **Eco-Epidemiology, Genotype Susceptibility**

INTRODUCTION

Banana (*Musa* spp.) is one of the most significant fruit crops worldwide, belonging to the family Musaceae. It thrives predominantly in tropical and subtropical climates and is extensively cultivated for its nutritional, economic, and social value. Botanically, the banana is a perennial herbaceous plant, characterized by its robust pseudostem, large broad leaves, and clusters of elongated fruits that originate from the plant's female inflorescence.[1,2]

Most cultivated bananas are sterile triploids, classified into genome groups such as AAA, AAB, and ABB, and are propagated through vegetative means, including suckers and tissue culture techniques, rather than through seeds. The fruit is a rich source of carbohydrates, dietary fiber, and essential micronutrients such as potassium, vitamin C, and vitamin B6, making it a vital component of the diet in many parts of the world.

In developing countries, banana cultivation plays a dual role—it serves both as a primary food crop and a crucial source of income for millions of smallholder farmers. In the Indian agricultural landscape, banana stands among the top-ranking fruit crops in terms of production volume and area under cultivation. States such as Maharashtra, Tamil Nadu, Bihar, and Kerala are among the leading producers.[3,4] Beyond its contribution to the national economy, banana farming also plays a pivotal role in ensuring food security, nutrition, and rural employment, especially in regions with limited crop diversity or income alternatives.

*Corresponding author :
Phone : 9006991000
E-mail : prf.arunkumar@gmail.com

The banana leaf and fruit scarring beetle (*Basilepta subcostatum* Jacoby) is a significant insect pest belonging to the Coleoptera order and Chrysomelidae family, known for inflicting serious damage to banana plants. This small beetle, often exhibiting a shiny metallic appearance, feeds aggressively on the surface tissues of banana leaves, pseudostems, and especially developing fruits. Adult beetles are responsible for producing distinctive linear scars and feeding abrasions on the fruit peel, which not only degrade the aesthetic and commercial value of the bananas but also make the fruits more vulnerable to secondary microbial infections.[5,6]

The immature stages (larvae) typically remain in the soil, where they feed on decaying organic matter or root tissues, while the adult beetles are found on the aerial parts of the plant. The dual life-stage behavior complicates detection and timely control. This pest is particularly prevalent in humid and warm climates, especially in the banana-growing belts of Eastern India, including Saharsa and adjoining districts of Bihar, where it is emerging as a major constraint to banana productivity and quality.7

Outbreaks tend to intensify during the post-monsoon season, when the environmental conditions favor beetle multiplication and activity. Despite the significant economic losses associated with this pest, its management remains difficult due to its sporadic emergence, cryptic habits, and lack of targeted control strategies. Hence, there is a pressing need to investigate the population dynamics, seasonal abundance, and varietal susceptibility of this pest in local agro-ecological conditions to design more sustainable, eco-friendly, and genotype-specific management approaches.[8]

Despite its importance, little is known about its seasonal behavior and host preference between cultivated and wild banana genotypes in North Bihar, especially Saharsa. This study addresses these gaps by investigating beetle population dynamics and comparative damage across genotypes.

MATERIALS & METHODS

Experimental Site and Period

The present investigation was carried out over a period of one year, from April 2022 to March 2023, at the Banana Research Farm located in Saharsa district, Bihar (latitude 25.88°N, longitude 86.60°E). This region lies within the subtropical agro-climatic zone of Eastern India and is well-known for extensive banana cultivation under both irrigated and rainfed conditions. The site experiences distinct seasonal variations, with hot summers, high humidity during the monsoon, and mild winters—conditions that are highly conducive to the activity and proliferation of banana scarring beetles (Basilepta subcostatum). The experimental location provided a representative agro-ecological setting to study the seasonal population trends, host plant interactions, and damage patterns of the pest under natural field conditions. The data collected during this time frame formed the basis for evaluating pest dynamics across local and wild banana genotypes, offering insights into environmentally suitable and host-specific management strategies..

Genotypes Studied

- Local cultivar: *Musa paradisiaca* cv. 'Malbhog'
- Wild genotype: *Musa balbisiana* (wild type)

Data Collection

To comprehensively assess the population dynamics and damage potential of the banana scarring beetle (*Basilepta subcostatum*), systematic data collection was carried out throughout the study period (April 2022 to March 2023) at the Banana Research Farm, Saharsa. The methodology was designed to capture both temporal and genotype-specific variations in beetle activity and associated plant damage under field conditions. The following procedures were employed:

- **Weekly Beetle Counts:**

For each banana genotype under observation-namely, the local cultivar *Musa paradisiaca* cv. 'Malbhog' and the wild species *Musa balbisiana*- 10 plants were randomly selected from each plot. On these plants, visual counts of adult beetles were recorded once every week. Observations were made during the early morning hours (7:00–9:00 a.m.) when beetle activity was most prominent and disturbances minimal. Care was taken to inspect the upper and lower leaf surfaces, pseudostems, and developing fruits, where beetles typically aggregate.

- **Leaf and Fruit Damage Assessment:**

The extent of feeding damage caused by adult beetles was recorded for both leaves and fruits using a standardized visual rating scale ranging from 0 to 5, where:

- 0 = No visible damage
- 1 = Very mild damage (1–10% surface affected)
- 2 = Mild damage (11–25%)
- 3 = Moderate damage (26–50%)
- 4 = Severe damage (51–75%)
- 5 = Very severe damage (>75%)

The scores were averaged weekly to monitor the severity of infestation and to facilitate comparison between the susceptible local and the more tolerant wild genotypes.

• **Recording of Meteorological Data:**

To understand the influence of climatic factors on beetle population fluctuations, daily weather data-including temperature (°C), relative humidity (%), and rainfall (mm)-were obtained from the nearby meteorological observatory. These parameters were aggregated on a weekly basis and analyzed for correlation with beetle abundance and damage intensity. Particular attention was paid to the post-monsoon period, during which beetle activity typically intensifies in the region due to favorable moisture and temperature levels.

Statistical Analysis

To interpret the variations in beetle population and the associated damage on different banana genotypes across various seasons, rigorous statistical tools were employed using SPSS version 25.0.

- Two-way Analysis of Variance (ANOVA):A two-way ANOVA was conducted to assess the interaction effects of two independent variables:

1. Banana genotype (*Malbhog* and *Musa balbisiana*)

2. Seasonal variation (Pre-monsoon: April–June, Monsoon: July–September, Post-monsoon: October–December, Winter: January–March)

The analysis revealed a highly significant difference ($p < 0.01$) in mean beetle population per plant between the two genotypes and among the four seasonal periods. Specifically:

• The local cultivar 'Malbhog' recorded the highest mean beetle count during the post-monsoon season (7.6 ± 0.4 beetles/plant).

• In contrast, the wild genotype '*Musa balbisiana'* showed significantly lower populations across all seasons, with a peak of only 3.0 ± 0.2 beetles/plant in the same period.

• Interaction effects (genotype × season) were also statistically significant, indicating that the seasonal influence on beetle abundance varied between the two genotypes.

Key findings include:

• A strong positive correlation ($r = 0.72$) between relative humidity and beetle population, suggesting that high humidity levels favor beetle activity and survival.

• A moderate positive correlation ($r = 0.58$) with rainfall, particularly evident during the monsoon and post-monsoon periods.

• A negative correlation ($r = –0.45$) with maximum temperature, indicating a decline in beetle activity when temperatures exceeded 35°C.

These statistical insights confirm that beetle infestation levels are significantly influenced by genotype susceptibility and seasonal environmental conditions. The wild banana genotype demonstrated inherent tolerance, while the local cultivar was more vulnerable, especially under humid post-monsoon conditions. Such findings underscore the importance of integrating genotype selection and climate-based monitoring into banana pest management strategies.

RESULTS & DISCUSSION

Population Dynamics of *Basilepta subcostatum*

The study revealed distinct seasonal trends in the population dynamics of the banana leaf and fruit scarring beetle, *Basilepta subcostatum*, across the experimental period from April 2022 to March 2023. A gradual increase in beetle activity was observed from the pre-monsoon season, with a pronounced peak during the post-monsoon months of September to October.

The highest mean beetle population was recorded in September, when environmental conditions—moderate temperatures (28–32°C) and high relative humidity (70–85%)—were favorable for beetle reproduction and feeding activity. In contrast, population levels declined sharply during the cooler and drier winter months (December–February), reaching the lowest values in March.

Month	Avg. Beetles/Plant (Malbhog)	Avg. Beetles/Plant (*Musa balbisiana*)
June	4.2 ± 0.3	1.3 ± 0.2
September	7.6 ± 0.4	3.0 ± 0.2
December	2.1 ± 0.3	0.6 ± 0.1
March	1.5 ± 0.2	0.4 ± 0.1

The results corroborate earlier reports that *B. subcostatum* thrives under moist and warm conditions. Its activity declines as temperatures drop and humidity decreases, which likely suppresses mating, oviposition, and foraging behaviors.

Adult beetles on Malbhog created distinct longitudinal scars and feeding abrasions on the fruits and young leaves, severely affecting fruit appearance and potentially reducing market value by over 40%. In contrast, the wild genotype experienced only minor feeding scratches, possibly due to differences in leaf texture, cuticle thickness, or chemical defenses.

These findings indicate that host plant resistance plays a vital role in pest dynamics. The genetic traits of *Musa balbisiana* may offer promising leads for developing resistant hybrids or varieties through future breeding programs.

Influence of Weather Parameters

Correlation analysis between beetle counts and climatic variables provided further insight into environmental influences on pest behavior:

- A strong positive correlation (r = 0.72) was recorded between relative humidity and beetle population, confirming the insect's preference for humid conditions.
- Rainfall also showed a moderate positive correlation (r = 0.58), likely aiding in the survival and emergence of adults from the soil.
- Conversely, high temperatures (>35°C) negatively impacted beetle activity, with a correlation coefficient of –0.45, indicating thermal thresholds beyond which beetle physiology or mobility may be impaired.

These relationships underscore the importance of climate-informed pest surveillance, particularly during the post-monsoon window, when interventions could be most effective.

CONCLUSION

The study of the banana leaf and fruit scarring beetle (*Basilepta subcostatum*) in the banana-growing region of Saharsa, Bihar, reveals key insights into the seasonal population dynamics, host plant interactions, and damage patterns of this pest. The findings indicate that beetle populations exhibit distinct seasonal fluctuations, with the highest abundance occurring during the post-monsoon period, when environmental conditions—particularly high humidity—are most favorable for their activity. The local cultivar *Musa paradisiaca* cv. 'Malbhog' demonstrated significant susceptibility to beetle infestation, with extensive leaf and fruit damage, leading to potential economic losses. In contrast, the wild genotype Musa balbisiana displayed remarkable resistance, showing fewer beetles and minimal damage. Environmental factors, particularly relative humidity and rainfall, were found to positively correlate with beetle population dynamics, emphasizing the importance of climate-based pest surveillance and control. The study highlights the potential of wild banana genotypes, such as *Musa balbisiana*, as a source of genetic resistance, offering promising prospects for breeding programs aimed at developing pest-resistant banana varieties. In conclusion, the integration of resistant genotypes, climate-informed pest management strategies, and early interventions during the post-monsoon surge are essential for mitigating the impact of *Basilepta subcostatum* on banana crops. Future research should explore the underlying mechanisms of resistance in wild bananas and evaluate eco-friendly control options, such as botanical repellents and biological control agents, to ensure sustainable and effective pest management in banana cultivation.

REFERENCES

1. **FAO. 2021.** Banana Market Review: FAO.
2. **Rajan, S. & Singh, H.P. 2011.** Banana Breeding in India: Current Status and Future Strategies. ICAR.
3. **Sharma, G. & Tewari, G.C. 2018.** Insect pests of banana and their management. *Journal of Entomological Research*, **42(2):** 165–172.
4. **Shyam Babu Sah, R.N. Gupta, Santosh Kumar, Tamoghna Saha and B.B. Singh. 2022.** Seasonal incidence of banana scarring beetle in organic banana in Bihar. *Annals of Plant Protection Sciences*, **30(1) :** 54-60
5. **Anshu Prabha, Arun Kumar & Bhuwan Bhaskar Mishra. 2021.** Eco-friendly management of banana scarring beetle, *Basilepta subcostatum* (Jacoby) in North Bihar. *Biospectra*, **16(1):**1-4.
6. **Pinku Paul, Sukhen Chandra Das, Saurav Saha and S Uma. 2020.** Studies on population dynamics and damage potential of banana leaf and fruit scarring beetle, *Basilepta subcostatum* Jacoby on local and wild banana genotypes of Tripura. *Journal of Entomology and Zoology Studies*, **8(4):**2020-2026
7. **Padmanaban, B., Kannan, M., Uma, S., Saraswathi, M.S., Backiyarani, S. and Ashif, K., 2020.** Field evaluation and *in vivo* screening of *Musa germplasm* against banana stem weevil, *Odoiporus longicollis*. *J Entomol Zool Stud*, **8(1):** 290-296.
8. **IPCC. 2021.** Climate Change 2021: The Physical Science Basis. Intergovernmental Panel on Climate Change.

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